Gene expression profiling in whole blood from ankylosing spondylitis patients and control groups using the R programming language

I. Introduction

Ankylosing spondylitis (AS) is a form of immune system failure in recognizing its own body tissues, which can trigger chronic inflammation, especially in the spine and sacroiliac joints. Although the genetic correlation between AS and the HLA-B27 allele has been well documented, the complexity of its pathogenesis indicates the involvement of other, broader molecular mechanisms. Therefore, exploring gene expression profiles is crucial for mapping inflammatory pathways and identifying potential biomarkers that drive the progression of this disease.

With the increasing availability of public genomic data on Gene Expression Omnibus (GEO), bioinformatics analysis has become the new standard in researching complex diseases. Unlike conventional manual approaches, this study applies a workflow based on the R programming language to perform differential expression analysis on whole blood samples from US patients.

The use of code-based computing not only improves precision in large-scale data processing, but also ensures more structured and reproducible analysis results. Through this approach, it is hoped that systemic molecular changes underlying inflammatory conditions in AS patients can be identified more comprehensively than with simple analysis methods.

II. Methods

A. Dataset

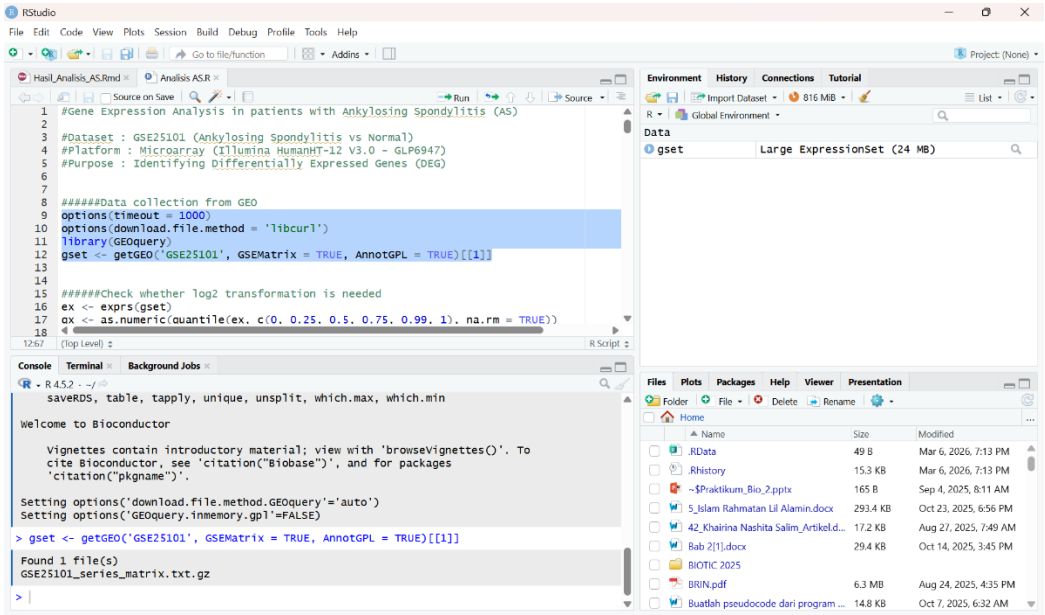
This study utilizes secondary gene expression data accessed through the Gene Expression Omnibus (GEO) with access number GSE25101. This dataset includes transcriptomic profiles from 32 whole blood samples processed using the Illumina HumanHT-12 V3.0 expression beadchip (GLP6947) microarray platform. Unlike previous approaches that used web-based tools (GEO2R), raw data from this platform was processed in an integrated manner through the R programming environment to ensure more precise quality control and normalization.

B. The R Workflow

i. Metadata acquisition

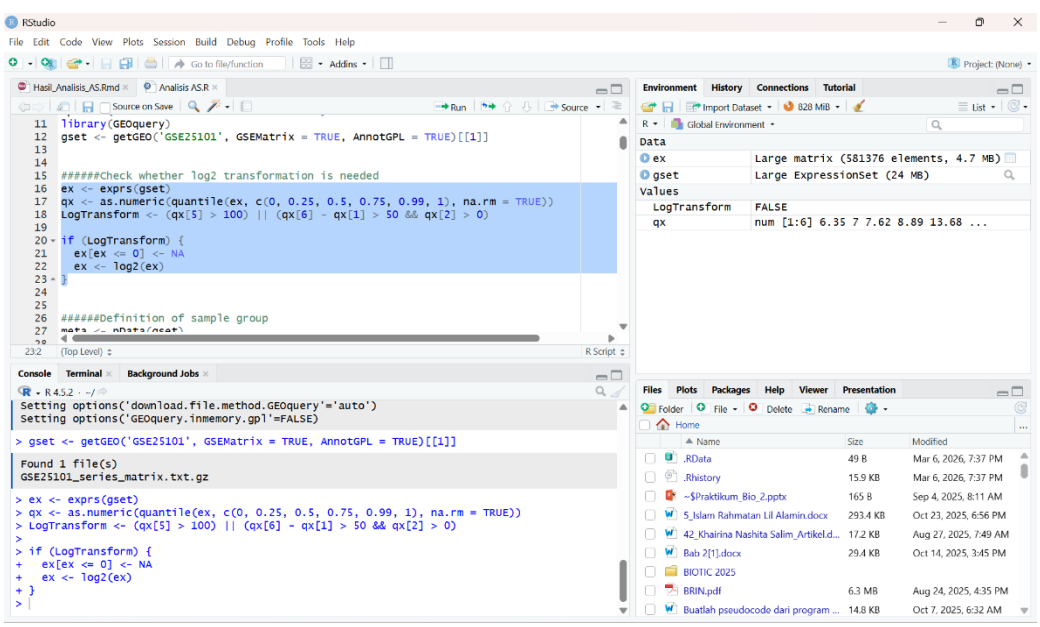
Transcriptomic analysis in this study was conducted entirely using the R programming language version 4.5.2 in the RStudio environment. This approach was chosen to ensure the measurability and reproducibility of all stages of analysis. The initial stage began with the development of a script for programmatic data acquisition. Unlike manual downloads,

metadata and raw expression matrices were extracted directly from public databases using functions in the GEOquery library. This process enabled seamless integration of expression data with sample clinical information without the risk of data manipulation errors.



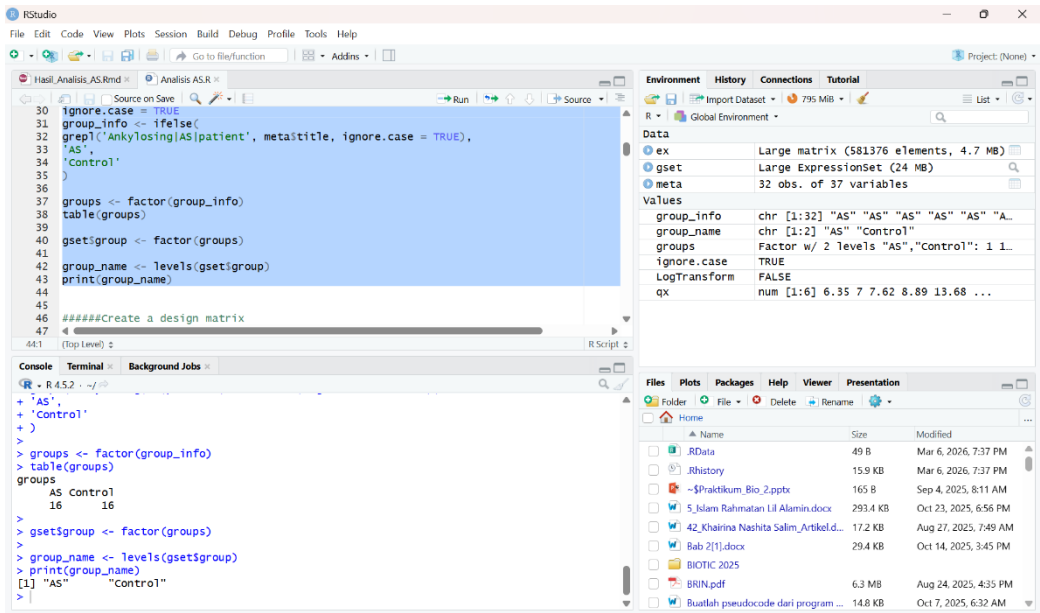
ii. Conditional logarithmic transformation

To ensure that the data has a normal distribution and to stabilize the variance, the data scale is checked using quantile values. The R script is designed to automatically detect whether the expression matrix has a large range of values. If detected, a log2 transformation will be applied. This step is important to minimize the impact of outliers.



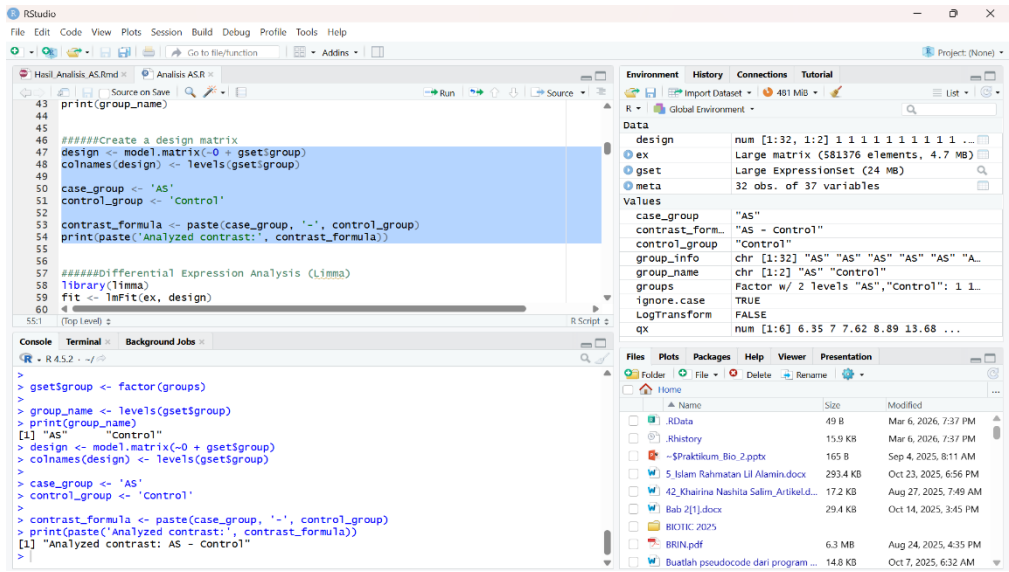
iii. Definition of sample groups

Metadata information from GEO was processed to separate samples into two groups: Patients (US) and Controls. This grouping was done through pattern matching on sample titles using the grepl function. As a result, each samples was given a consistent factor label for use in statistical modeling.



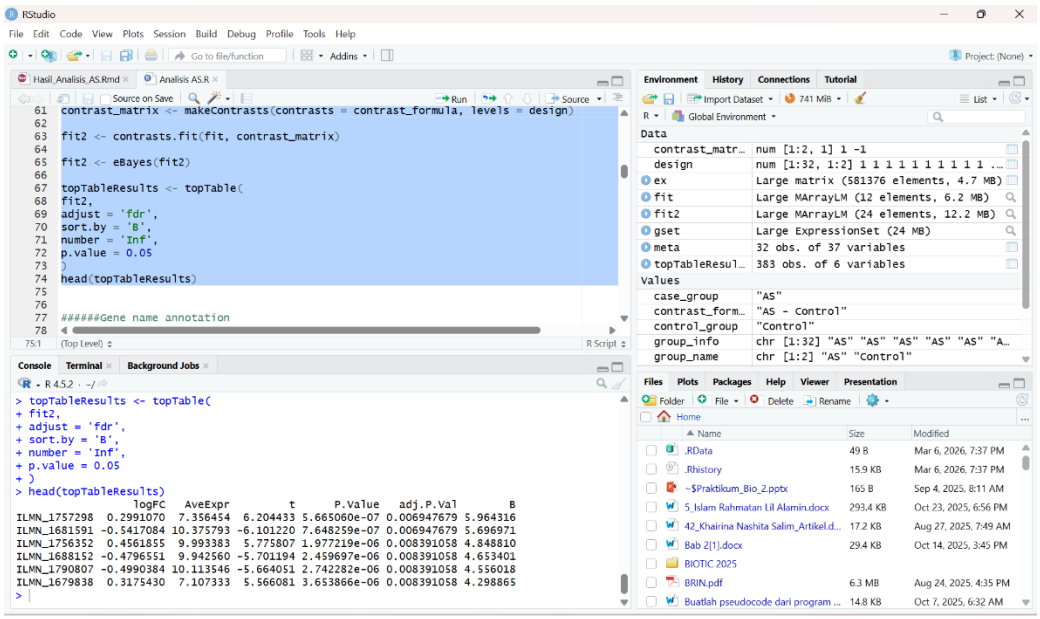
iv. Matrix design and contrast formula

To compare gene expression differences between groups, a matrix design was created using the model.matrix function. This matrix represents the experimental structure at the intercept (~0), followed by the definition of contrasts (AS - Control). This formula forms the basis for the algorithm to calculate which genes are upregulated or downregulated in AS compared to healthy individuals.



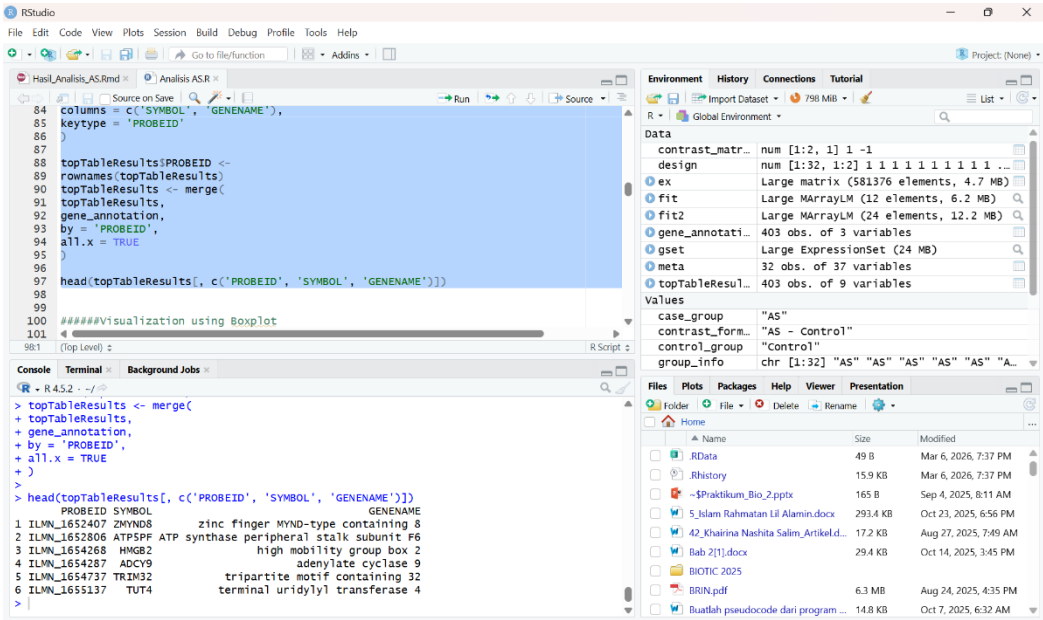
v. Differential expression analysis (limma)

Identification of differentially expressed genes was performed using the limma package with a linear model approach. The process began by integrating the expression matrix into the design matrix and creating a contrast matrix (AS - Control) to compare the patient group with the control group. Next, the Empirical Bayes (eBayes) method is applied to stabilize variance estimates and increase the statistical power of the test. The results will be filtered using the False Discovery Rate (FDR) with a significance threshold of adj.P.Val < 0.05. All genes that meet these criteria will be extracted into a results table for further analysis and visualization.



vi. Gene name annotation

Since the initial analysis results are still in the form of Illumina platform-specific probe IDs, the gene annotation stage is carried out using the illuminaHumanv3.db library. This process maps each probe ID to a gene symbol and standard full gene name through the AnnotationDbi::select function. The mapping results are then combined back into the main results table to facilitate the biological identification of differentially expressed genes.

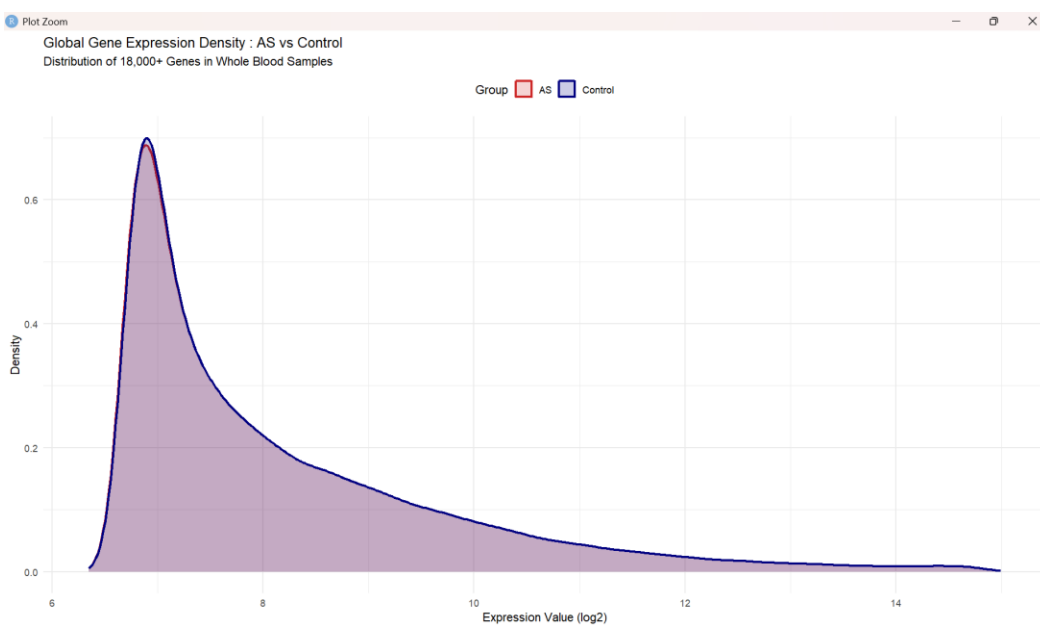


III. Results and Interpretation

A. Visualization results

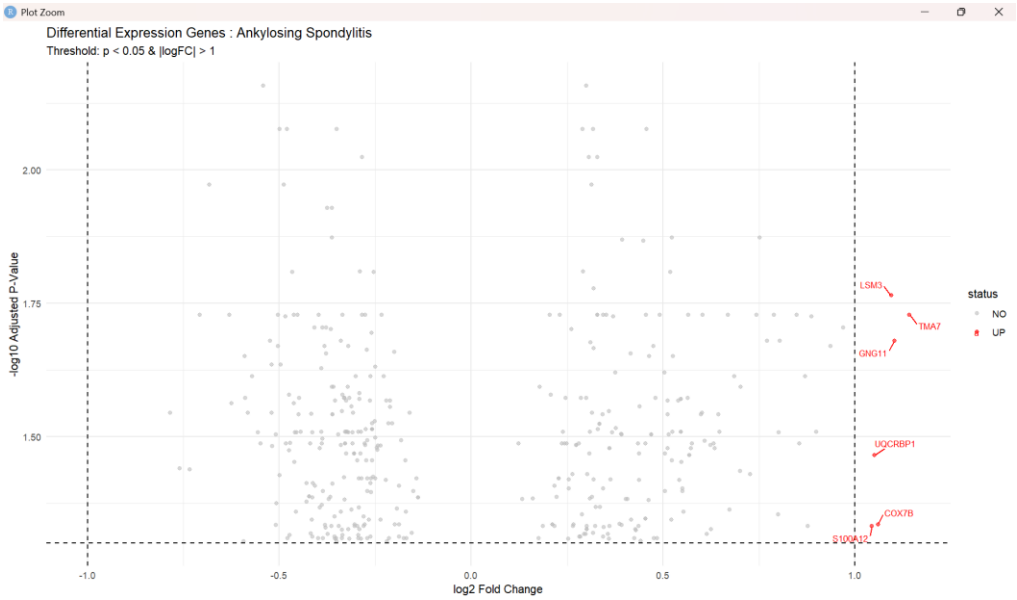
i. Density plot

Visualization using density plots shows that after log2 transformation, the distribution of gene expression intensities in 32 samples has a uniform profile. This confirms that there is no significant technical bias between samples, making the data suitable for further analysis.



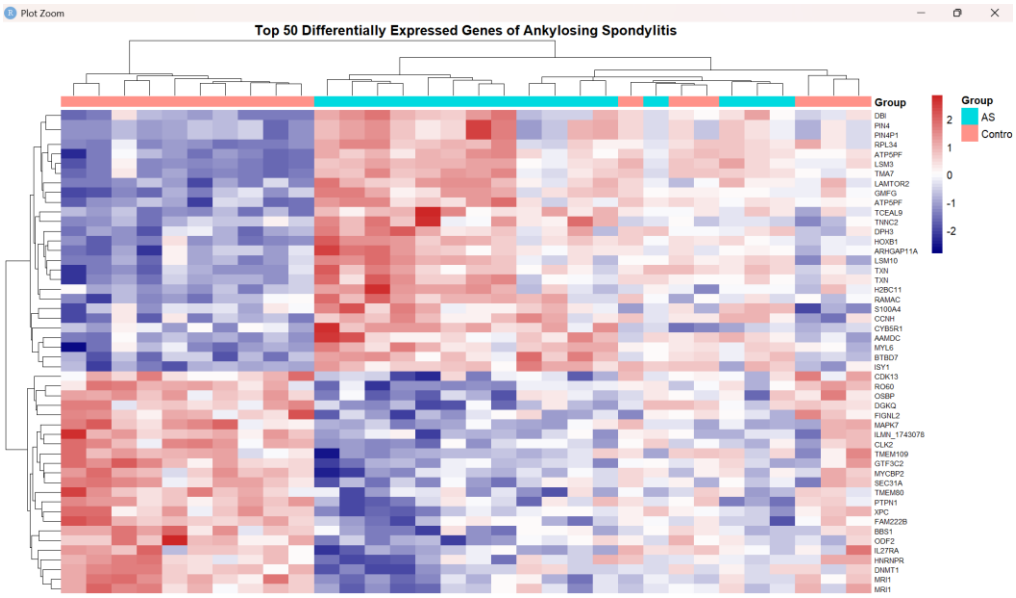
ii. Volcano plot

Based on statistical analysis results, the relationship between the level of expression change (logFC) and significance (adj.P.Val) is mapped in a Volcano plot. With a threshold of |logFC| > 1 and $p < 0.05$, there is a distribution of genes that are significantly up-regulated and down-regulated in patients with ankylosing spondylitis.



iii. Heatmap

The visualization results show a clear separation between the AS and control groups, where the gene expression profile in AS tends to show a uniform activation trend in certain molecular pathways. Although only six genes meet the very strict significance criteria, this heatmap provides a comprehensive overview of the contrasting molecular markers between AS and the control group.



B. Biological interpretation

The results of differential expression analysis showed that there were six main genes that were significantly upregulated in AS patients. These genes can be categorized into the following mechanisms:

1). Markers of acute inflammation and immune activation

S100A12 is the most crucial finding in the gene list because this gene encodes a calcium-binding protein released by neutrophils during the inflammatory process. Its increase in AS patients is a strong indicator of neutrophil activation and inflammation. This protein triggers the formation of pro-inflammatory cytokine cascades that contribute to damage to joint and spinal tissue.

2). Energetic metabolism and oxidative stress

Increased metabolic genes (COX7B and UQCRBP1) indicate that immune cells in AS patients are in a state of high metabolism to support sustained inflammatory activity. This is also often associated with the production of reactive oxygen species, which exacerbates oxidative stress in the area of inflammation.

3). RNA regulation and protein synthesis

Collectively, the increase in LSM3, TMA7, and GNG11 indicates that cells are undergoing massive protein remodeling, most likely to produce inflammatory mediators or respond to chronic cytokine signals in the bodies of AS patients.

IV. Enrichment Analysis

Although differential expression analysis successfully identified key genes that are highly relevant biologically, such as S100A12, GO and KEGG enrichment analyses did not produce statistically significant pathways at an adj.P.Val threshold < 0.05. This is most likely due to:

1). A very limited number of genes

The significance criteria were strict, and only six major genes passed the filtration, thus failing to reach the minimum threshold for functional accumulation in the standard database.

2). Characteristics of the disease

As a disease with a specific pathogenesis, some of the expressed genes may be involved in pathways that have not been fully annotated in public databases.

Therefore, interpretation of the results focused on the individual roles of significant genes that have been shown to have a strong correlation with inflammatory mechanisms in ankylosing spondylitis.

V. Conclusion

This study successfully mapped the gene expression profile in patients with Ankylosing Spondylitis (AS) using a bioinformatics approach based on the R programming language. Through a systematic pipeline, six key genes were found to be significantly upregulated, with S100A12 as the most prominent inflammatory indicator showing neutrophil activation. These findings are supported by heatmap results showing a clear separation of molecular patterns between the AS patient group and the control group.

Methodologically, the transition from manual analysis to the use of R has been shown to improve the precision, efficiency, and reproducibility of analysis results. Overall, the integration of public genomic data and appropriate computing can provide a deeper understanding of the molecular mechanisms behind chronic inflammation in the pathogenesis of ankylosing spondylitis.